

DECODE LABS

IoT DEPARTMENT | INTERNSHIP PROGRAM

WEEK 2 PROJECT

Project 2: Automated Irrigation Controller (Actuator Logic & Closed-Loop Feedback System)

Analog Sensing • Relay Actuation • ADC Logic

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Project 2: Automated Irrigation Controller (Actuator Logic)

1. Project Goal

Build an automated, closed-loop feedback system that activates mechanical hardware based on real-time environmental input. The system continuously monitors soil moisture using an analog sensor, applies threshold logic to decide whether the soil is too dry, and actuates a relay-controlled 5V water pump to irrigate the soil automatically — without any manual intervention.

2. Key Requirements

- Read analog input variables from a Soil Moisture Sensor.
- Design an explicit threshold logic gate to evaluate if the soil is too dry.
- Trigger a digital output pin to switch a 5V relay module on/off, simulating a water pump actuator.
- Maintain a continuous closed-loop monitoring cycle (read → evaluate → act → repeat).

3. Key Skills Demonstrated

- **Analog-to-Digital Conversion (ADC)** — converting the sensor's analog voltage into a usable digital moisture value (0–1023).
- **Electronic Relay Mechanics** — using a low-current digital signal to safely switch a higher-current pump circuit.
- **Closed-Loop Threshold Logic** — comparing live sensor data against a defined threshold to make an autonomous decision.

4. Components Used

Component	Role in Circuit	Connected To
Arduino Uno	Microcontroller / Brain of the system	All modules
Soil Moisture Sensor	Provides analog moisture reading (AOUT)	A0 (analog input)
5V Relay Module	Digitally switches the high-current pump circuit	D7 (digital output)
5V DC Water Pump	Actuator — physically pumps water to the soil	Relay NO / COM terminals
External 5V DC Supply	Powers the pump load independently of the Arduino	Relay COM terminal
Breadboard	Distributes shared 5V and GND rails	Arduino + Sensor + Relay

5. Circuit Diagram (Tinkercad-Style Simulation)

The circuit below was designed and simulated in a Tinkercad-style breadboard layout. The Arduino Uno supplies 5V/GND to the breadboard's power rails, which distribute power to the soil moisture sensor and relay module. The sensor's analog output feeds pin A0, while a digital control signal from pin D7 drives the relay's IN pin. The relay's COM and NO terminals switch the external pump power circuit on and off.

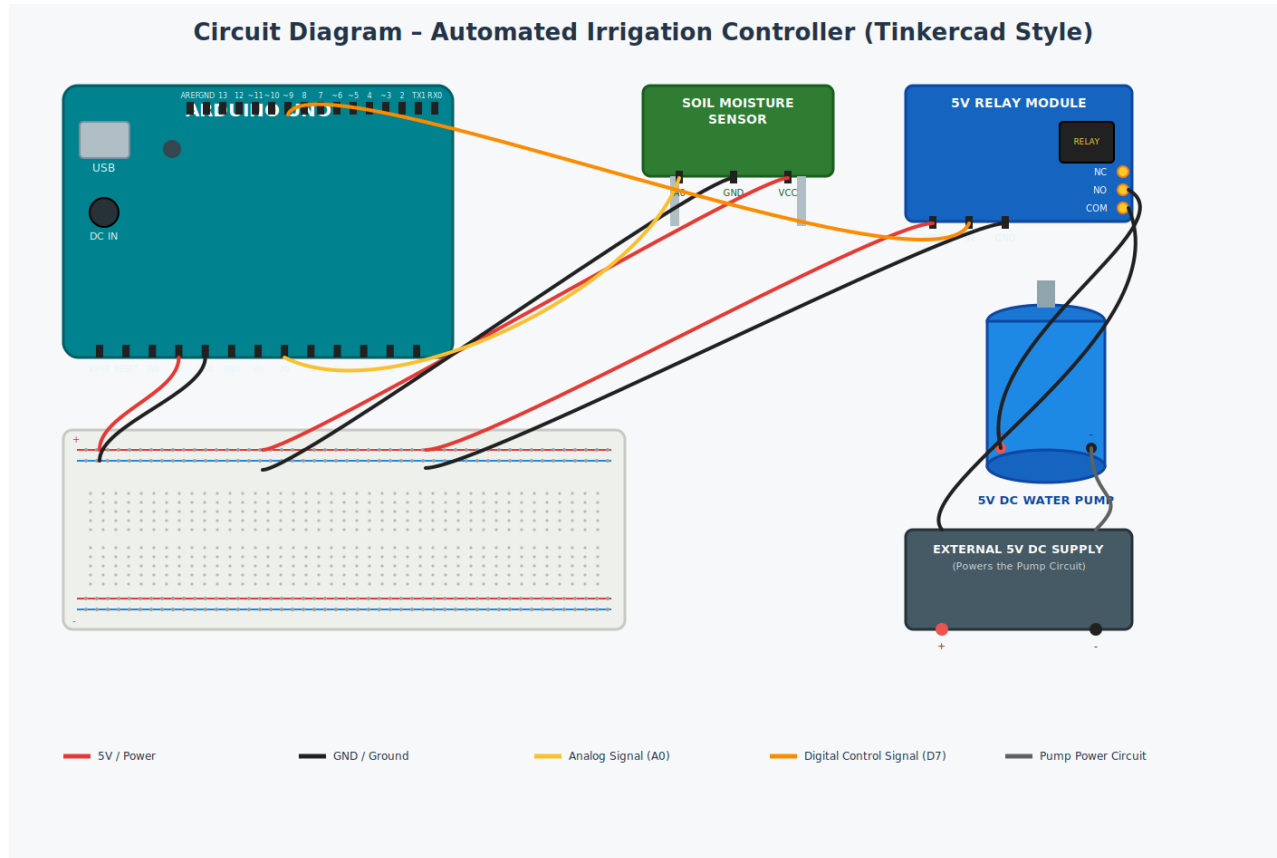


Figure 1: Full circuit wiring — Arduino Uno, Soil Moisture Sensor, Relay Module, and 5V Water Pump.

5.1 Wiring Summary

Signal	From	To	Wire Color
Power (5V)	Arduino 5V pin	Breadboard (+) rail	Red
Ground	Arduino GND pin	Breadboard (-) rail	Black
Sensor Power	Breadboard (+) rail	Sensor VCC	Red
Sensor Ground	Breadboard (-) rail	Sensor GND	Black
Analog Signal	Sensor AO/UT	Arduino A0	Yellow
Relay Power	Breadboard (+) / (-) rails	Relay VCC / GND	Red / Black
Relay Control	Arduino D7	Relay IN	Orange

Pump Circuit	External 5V Supply → Relay COM/NO Pump	Relay NC → Supply (-)	Black (thick)
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6. Closed-Loop Threshold Logic

The controller repeatedly executes a read-evaluate-act cycle. On every loop iteration, the Arduino reads the raw analog value from the soil sensor, converts it into a moisture percentage, and compares it against a pre-defined dry threshold. This explicit threshold gate is the decision point that determines whether the relay (and therefore the pump) should be switched on.

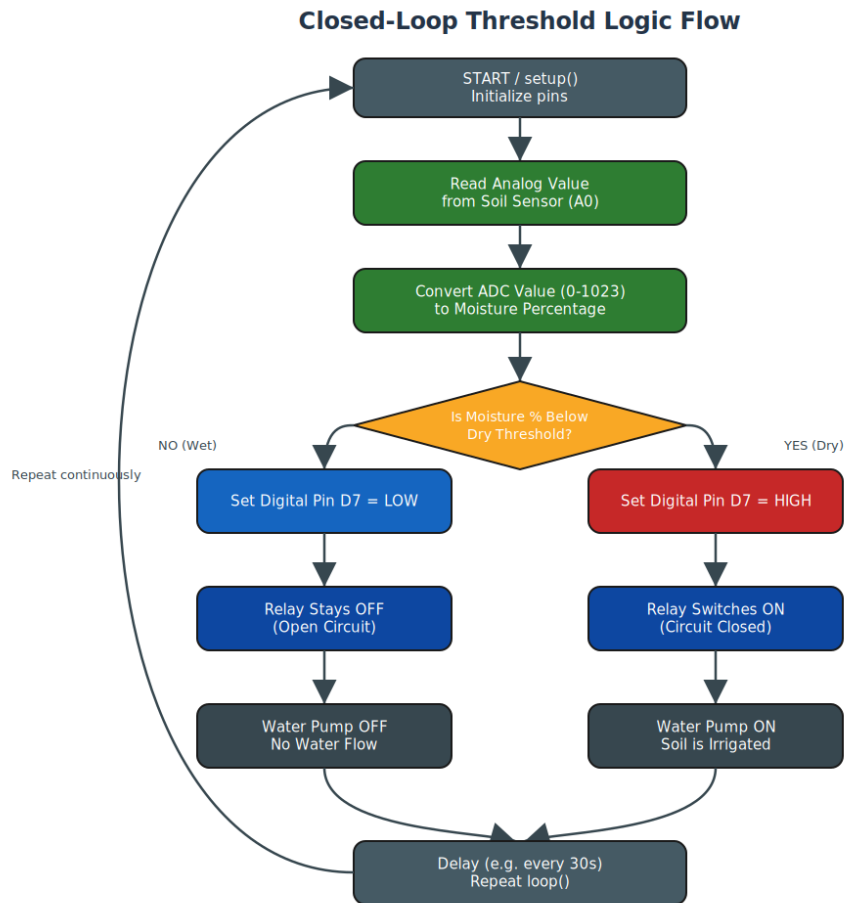


Figure 2: Closed-loop decision flow implemented inside `loop()`.

7. Arduino Sketch (Actuator Logic)

The sketch below implements the ADC read, the threshold logic gate, and the relay actuation described above. The dry threshold and sensor calibration values were derived from typical resistive soil moisture sensor behavior (dry soil reads a high analog value close to 1023; saturated soil reads a low value close to 0), and are exposed as constants so they can be re-calibrated to the specific sensor and soil used.

```
// =====
// Project 2: Automated Irrigation Controller (Actuator Logic)
// Board: Arduino Uno
// =====

const int SOIL_SENSOR_PIN = A0;    // Analog input from Soil Moisture Sensor
const int RELAY_PIN       = 7;     // Digital output to Relay Module (IN)

// --- Calibration constants (adjust per sensor/soil) ---
const int DRY_ADC_VALUE = 1023;    // ADC reading in completely dry air
const int WET_ADC_VALUE = 300;     // ADC reading in fully saturated soil

// --- Threshold logic gate ---
const int MOISTURE_THRESHOLD_PERCENT = 40; // Below this % => soil is "dry"

// --- Timing ---
const unsigned long READ_INTERVAL_MS = 30000; // read every 30 seconds
unsigned long lastReadTime = 0;

void setup() {
  pinMode(RELAY_PIN, OUTPUT);
  digitalWrite(RELAY_PIN, LOW); // Ensure pump is OFF at startup
  Serial.begin(9600);
}

void loop() {
  unsigned long now = millis();

  if (now - lastReadTime >= READ_INTERVAL_MS) {
    lastReadTime = now;

    // 1. Read analog input (ADC)
    int rawValue = analogRead(SOIL_SENSOR_PIN);

    // 2. Convert raw ADC value to moisture percentage
    int moisturePercent = map(rawValue, DRY_ADC_VALUE, WET_ADC_VALUE, 0, 100);
    moisturePercent = constrain(moisturePercent, 0, 100);

    Serial.print("Raw ADC: ");
    Serial.print(rawValue);
    Serial.print(" | Moisture: ");
    Serial.print(moisturePercent);
    Serial.println(" %");

    // 3. Explicit threshold logic gate
    if (moisturePercent < MOISTURE_THRESHOLD_PERCENT) {
      // Soil is too dry -> activate relay -> pump ON
      digitalWrite(RELAY_PIN, HIGH);
      Serial.println("Status: DRY -> Relay ON -> Pump Watering Soil");
    }
  }
}
```

```
    } else {  
        // Soil moisture is sufficient -> deactivate relay -> pump OFF  
        digitalWrite(RELAY_PIN, LOW);  
        Serial.println("Status: WET -> Relay OFF -> Pump Idle");  
    }  
}  
}
```

8. How the Closed-Loop System Works

8.1 Analog-to-Digital Conversion (ADC)

The soil moisture sensor outputs a variable analog voltage (0–5V) that changes with the electrical resistance of the soil between its probes. The Arduino's onboard ADC samples this voltage and converts it into a 10-bit digital number ranging from 0 to 1023. Dry soil has higher resistance and produces a higher raw ADC value, while moist soil has lower resistance and produces a lower value. The `map()` function rescales this raw value into an intuitive 0–100% moisture reading using the two calibration constants `DRY_ADC_VALUE` and `WET_ADC_VALUE`.

8.2 Threshold Logic Gate

The core decision-making element of the system is a single explicit comparison: `moisturePercent < MOISTURE_THRESHOLD_PERCENT`. This threshold gate converts a continuous analog measurement into a binary actuation decision, forming the logical bridge between sensing and actuation. Setting the threshold too low risks under-watering; setting it too high risks over-watering, so the constant is intentionally exposed for easy recalibration.

8.3 Relay Mechanics and Actuation

The Arduino's digital pins cannot safely drive a water pump directly, since pumps draw far more current than a microcontroller pin can supply. Instead, digital pin D7 sends a small control signal (HIGH/LOW) to the relay module's IN pin. Internally, this energizes an electromagnetic coil that mechanically pulls a switch contact, connecting the relay's COM (common) terminal to its NO (normally open) terminal. This closes the separate, externally-powered high-current circuit that runs the pump — electrically isolating the low-power control logic from the high-power actuator circuit.

8.4 Closed-Loop Feedback

The system is described as “closed-loop” because it continuously re-reads the environment (soil moisture) after every actuation cycle rather than running the pump for a fixed, blind duration. As the pump adds water, the moisture percentage naturally rises; once it crosses back above the threshold, the same logic that turned the relay on will automatically turn it back off on the next read cycle — with no manual intervention required.

9. Testing Notes (Tinkercad Simulation)

- In the Tinkercad simulator, the Soil Moisture Sensor can be dragged to different simulated moisture levels to test both branches of the threshold logic.
- The Serial Monitor was used to verify the raw ADC value, computed moisture percentage, and relay state on every read cycle.
- The relay's click sound and onboard LED indicator confirm the switching action before the pump circuit is verified.

- Recommended next step: add a second, higher threshold (hysteresis) to prevent the relay from rapidly toggling near the boundary value.